

Tread Softly into The Future

The year is 2015. Technology has taken over the world. It's not the Terminator story yet, but we are almost there...

Ram Mohan Rao

It is a truism that every device is interconnected. They all recognise one another, they all understand each other. Unfortunately, they all put each other at risk. Welcome to the future!

Soon, viruses on your PCs will be passé. Spyware will be antiquated, and you will actually wish the 'good old days' were back. Of the several tech threats that lurk around the corner, these are the ones that stand high on the maximum disaster index.

Smartphones

Picture this: You are at your office, equipped with a newly configured WLAN (Wireless Local Area Network), and you receive an e-mail on your smartphone. Business as usual, right? It turns out the e-mail has a virus attachment. The attachment has a virus payload that reads and sends malicious e-mail and spam MMSes to all the contacts in your address book. The same, multi-faceted virus also propagates over any wireless network you are connected to at that moment. Your WLAN is now infected.

You don't realise you have a virus, and promptly sync your handheld with your PC as soon as you reach home. Once on the PC, the virus takes on a sinister role—part of the virus attaches itself to all digital devices connected to the PC, propagating via memory cards, synchronisations and the like. One part directly

attacks the host computer. One part uses your PC to spam; another launches a DoS attack.

It's strange that it hasn't happened already. The only well-documented viruses for wireless devices thus far are proof-of-concept ones—and now, as of August, we also have WinCE.Brador.a for Pocket PCs (See box "I Have Seen The Future, And I Am Scared").

Smartphone users routinely download ringtones and Java games. However, when it comes to anti-virus software, firewalls or spam filters for these devices, there aren't many, as of yet. This lack of software development should change with the rising popularity of mobile devices. Unfortunately, so will virus development. Thus far, virus developers have ignored such devices only because of the small numbers they can affect.

Tomorrow, however, there will be millions of smart phone users, and each user will have hundreds of contacts, with each contact just an e-mail, SMS or MMS away. Attack one phone, and you cause a domino effect that will sweep across the globe in a few hours. We switch off our computers, we barricade ourselves with firewalls and anti-virus software, but we rarely ever turn off our cell phones, or bother to think twice when we receive an SMS or an MMS. So how do we prevent this?

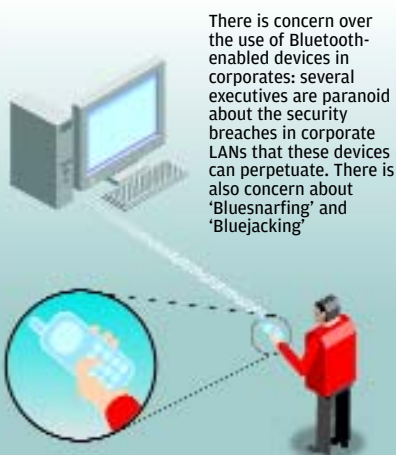
Take the Cabir virus, for example. It was sent to security firms for analysis; it wasn't released "into the wild". But as security firm F-Secure noted, the next attack might not be so civilised.

In 2003, the *New York Times* reported that the



Illustration: Mahesh Benkar

Chinks In The Wireless Armour



Infographic: Solomon Lewis